

Sai Keerthana Dulam

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Objective:

Fourth-year Computer Science undergraduate at NIT Patna with strong foundations in Machine Learning, SQL, and Data Analytics. Passionate about building data-driven solutions and interpretable AI systems to solve real-world problems.

Education:

National Institute of Technology, Patna B.Tech
in Computer Science and Engineering

Expected Graduation: 2027

CGPA (till 6th Semester): 7.88

Experience:

Internpe – **Machine Learning Intern**

Research:

An Explainable and Clinically Interpretable Machine Learning Framework for Ethical Cardiovascular Risk Prediction (*Manuscript Submitted to BMC Public Health, Springer Nature*)

- Submitted a research manuscript titled "*An Explainable and Clinically Interpretable Machine Learning Framework for Ethical Cardiovascular Risk Prediction*" to *BMC Public Health (Springer Nature)*; currently in the **Research Square In Review** process.

Technical Skills:

- Programming Languages: **Python (proficient)**, C++ (basic)
- Database and Query Languages: **SQL (MySQL)**
- Data Analysis and Visualization: **Microsoft Excel, Tableau** (In progress)
- Machine Learning: Supervised learning, feature engineering, model evaluation, cross-validation
- Libraries & Frameworks:
NumPy, Pandas, scikit-learn, PyTorch (basic)
- Maths for ML Linear algebra, probability & statistics, optimization intuition, and error analysis
- Tools & Platforms: Git, GitHub, VS Code, MySQL Workbench, Tableau
- Data Structures & Algorithms: Problem-solving (Basic DSA)
- Soft Skills: Problem-solving, teamwork, communication

Academic & Mini Projects:

1. **Explainable Machine Learning System for Cardiovascular Risk Assessment (Basic–Advanced Models)**

Tech Stack: Python, scikit-learn, Pandas, NumPy

- Built an end-to-end ML pipeline for **10-year cardiovascular risk prediction** using clinical data with preprocessing, feature engineering, and model evaluation.
- Implemented Logistic Regression and tree-based models with explainability techniques and designed a **tiered prediction framework (Basic → Middle → Advanced)**.

2. **Retail Sales Analytics using MySQL & Tableau**

Tech Stack: MySQL, Tableau, Python

- Built an end-to-end retail analytics pipeline on **9,994 sales records**, performing Python-based EDA and data quality analysis, MySQL integration, and **22 SQL queries** using CTEs, window functions, aggregations, and profitability analysis.
- Developed an **interactive Tableau dashboard** tracking **\$2.30M sales**,

\$286.4K profit, 12.47% margin, regional/category performance, loss-making sub-categories, and discount-profit trends with dynamic region filtering.

**3. Fake News Detection using Machine Learning
(Minor Project) Tech Stack: Python, scikit-learn,
TF-IDF, XGBoost**

- Developed an NLP-based fake news detection system using **TF-IDF** and machine learning models on multiple public datasets.
- Achieved **97.5% accuracy** with XGBoost by building an end-to-end pipeline for preprocessing, feature extraction, training, and evaluation.

**4. Python Mini-Projects (Code-in-Place, Stanford
University) Tech Stack: Python**

- Implemented multiple Python programs focusing on logic building, functions, and debugging.
- Developed strong foundations in clean coding and structured problem-solving.

Certifications:

1. Python Programming – Code-in-Place, Stanford University
1. NPTEL - The Joy of Computing Using Python (Silver Medal with 88%)
2. NPTEL – Human Computer Interaction (Gold Medal with 91%)

Positions of Responsibility:

- **Coordinator**, Total Chaos (Dance Club) & Recreation Club, NIT Patna — managed event coordination, teamwork, and execution.

Achievements & Extracurricular Activities:

- 3-Star Coder (Python) – HackerRank
- Active member of Google Developer Student Club (GDSC), NIT Patna
- Participated in an AI-domain Hackathon as part of a four-member team
- Represented NIT Patna in dance events; received certificate for outstanding performance

Interests: Applied mathematics for ML, explainable AI, quantitative problem-solving